POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

Video ID:	ec65d14e-4460-40fb-bc48-973ec5f4ae60
Total Frames:	2768
Frame Rate:	25.0 fps (assumed)
Duration:	110.7 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	117	✓
RT-DETR Detections	1086	✓
MedSAM2 Detections	641	✓
Consensus (3-Model Agreement)	5	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	5	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 447 frames)

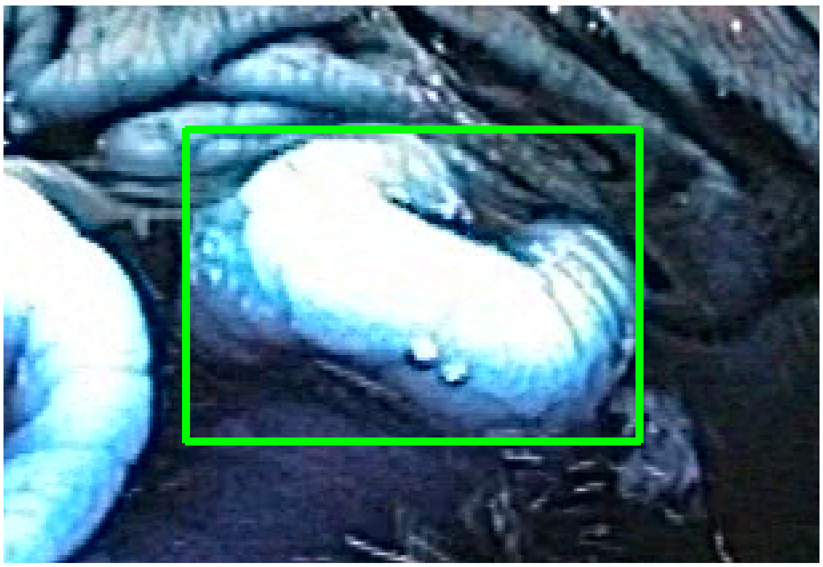
Feature	Value	Interpretation
Frame Index	946	Frame 946 of 2768
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 76.4%
Redness Score	0.001	Color-based vascularization
Relative Radius	58.6%	Polyp size as % of frame diagonal
Texture Score	0.361	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	68.3%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 56 frames)



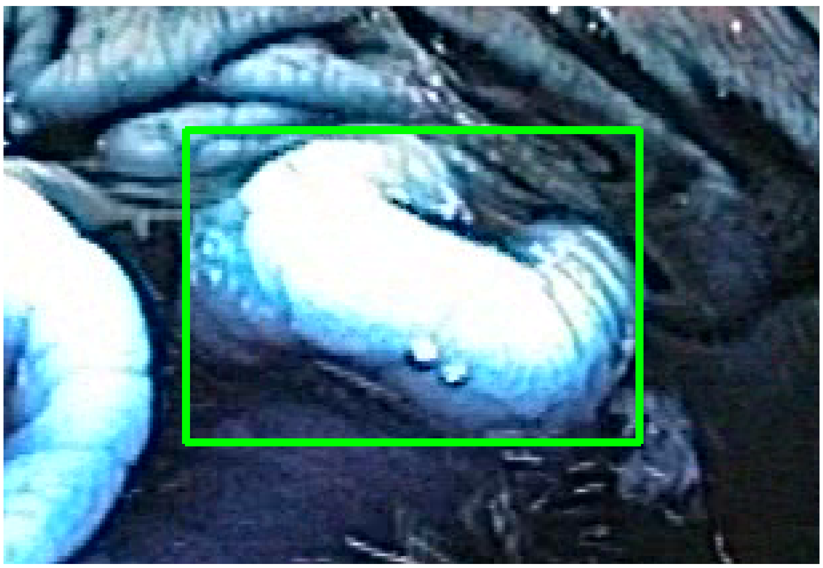
Feature	Value	Interpretation
Frame Index	597	Frame 597 of 2768
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 73.3%
Redness Score	0.001	Color-based vascularization
Relative Radius	28.4%	Polyp size as % of frame diagonal
Texture Score	0.247	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	71.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 56 frames)



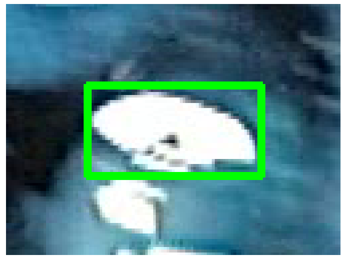
Feature	Value	Interpretation
Frame Index	597	Frame 597 of 2768
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 73.3%
Redness Score	0.001	Color-based vascularization
Relative Radius	28.4%	Polyp size as % of frame diagonal
Texture Score	0.247	Surface morphology
Vessel Visibility	0.000	Vascularization indicator

Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	68.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — LIFTED_POLYP [LOW_RISK] (Tracked across 56 frames)

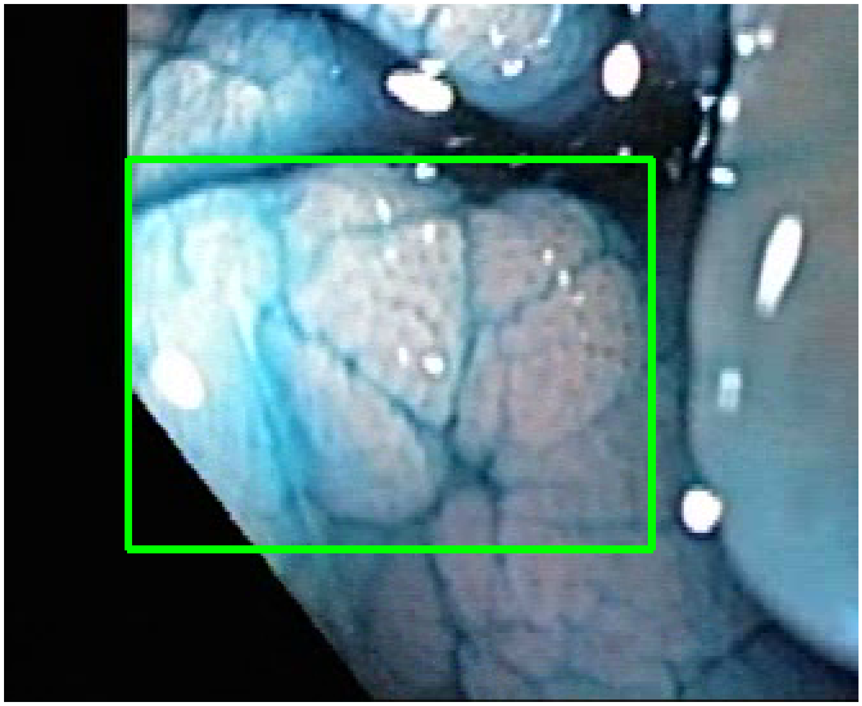


Feature	Value	Interpretation
Frame Index	1988	Frame 1988 of 2768
Detection Method	Detector Consensus (YOLO+RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 92.0%
Redness Score	0.000	Color-based vascularization
Relative Radius	7.2%	Polyp size as % of frame diagonal
Texture Score	1.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #5 — LIFTED_POLYP [LOW_RISK] (Tracked across 16 frames)



Feature	Value	Interpretation
Frame Index	1186	Frame 1186 of 2768
Detection Method	Detector Consensus (YOLO + RT-DETR agreement)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 67.8%
Redness Score	0.005	Color-based vascularization
Relative Radius	22.8%	Polyp size as % of frame diagonal
Texture Score	0.236	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	68.3%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 5 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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